

THIRD YEAR / FIRST SEMESTER

LECTURE 01

Introduction to Digital Communication

Systems, channels and sampling

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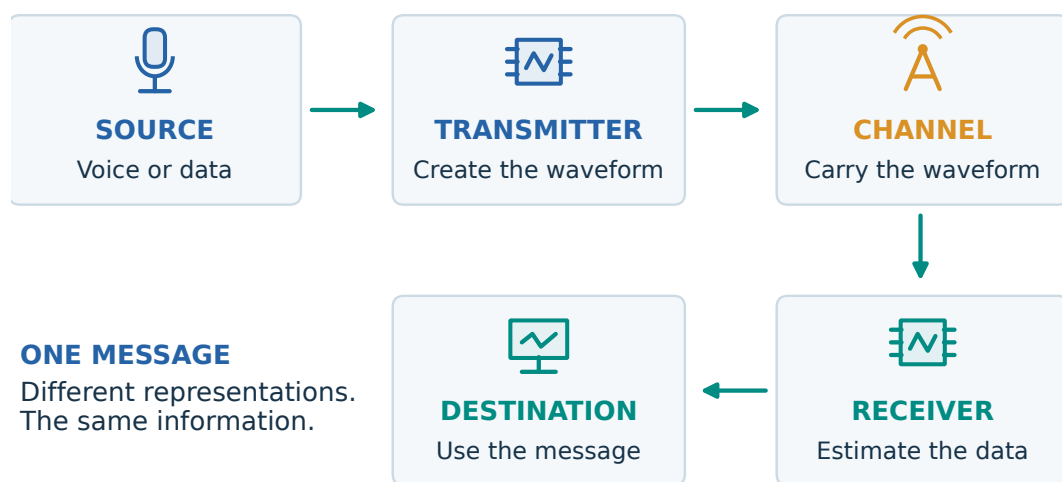


Figure 1.1: A message is prepared, transmitted and recovered.

What will you learn?

1. Identify the main parts of a digital communication system.
2. Explain sampling, quantization and binary encoding.
3. Compare wired and wireless channels.
4. Recognize attenuation, distortion and noise.
5. Find sampling rates and solve simple aliasing problems.

Remember. Read the explanation, follow the figure, then solve the example. The detailed sampling derivations come after the basic ideas and examples.

1.1 What is digital communication?

Definition: Digital communication

Digital communication sends information using a finite set of symbols. In a binary system, the information is represented by bits: 0 and 1.

A voice message begins as sound. A microphone changes the sound into a voltage. The system represents this voltage by numbers, sends those numbers, and uses them to recover the message. The receiver does not need an exact copy of every transmitted voltage. It must identify the correct symbols, then interpret their agreed meaning.

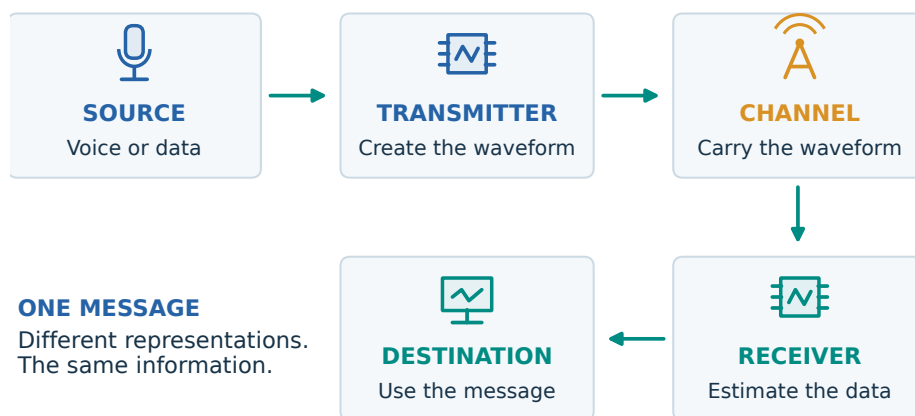


Figure 1.2: Follow the arrows from the source to the destination.

1. **Source.** Produces information, such as voice, text or a photograph.
2. **Transmitter.** Changes the information into a waveform suitable for the channel.
3. **Channel.** Carries the waveform through a cable, optical fiber or free space.
4. **Receiver.** Uses the received waveform to estimate the transmitted information.
5. **Destination.** Uses the recovered information, for example through a loudspeaker or a screen.

Remember. Bits are labels. The channel carries a physical waveform, such as a voltage, a light signal or a radio wave.

Quick question

In a voice call, what is the source? **Answer:** The speaker's message. The microphone converts the sound into an electrical signal.

1.2 From an analog signal to bits

Definition: Analog signal

An analog signal represents information using a continuously varying quantity, such as microphone voltage.

Before an analog signal can be represented by bits, it passes through three different operations.

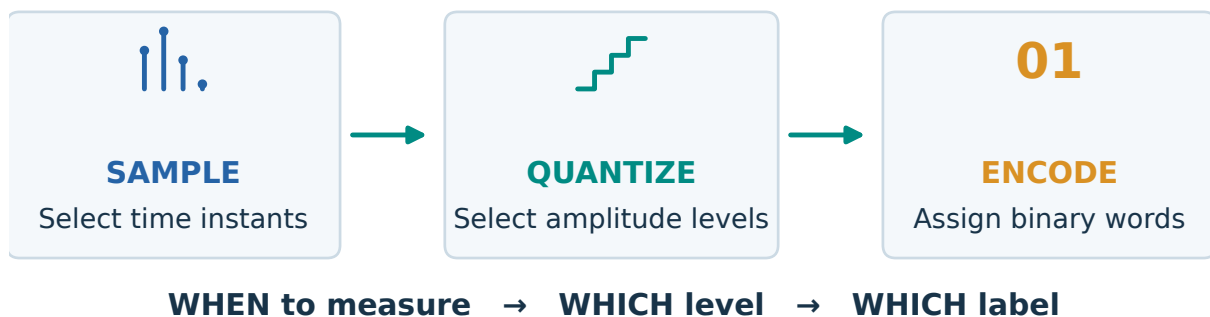


Figure 1.3: Sampling chooses times, quantization chooses levels, and encoding assigns labels.

1. **Sample.** Measure the voltage at regularly spaced times.
2. **Quantize.** Replace each measurement by the nearest allowed level.
3. **Encode.** Write the selected level as a binary word.

Together, these operations form an **analog-to-digital converter (ADC)**. Sampling preserves the measured value at a chosen time. Quantization introduces rounding. Encoding changes the label, not the chosen level.

Example 1

Question

A sample has value 0.62 V. The allowed levels are 0, 0.25, 0.50 and 0.75 V, with labels 00, 01, 10 and 11. What word is produced?

Solution

Given: Measured sample: 0.62 V. Four allowed levels.

1. **Find the nearest level.** The distances to 0.50 and 0.75 V are 0.12 and 0.13 V. Choose 0.50 V.
2. **Read its label.** The label of 0.50 V is 10.

Answer: The transmitted label for this sample is **10**.

1.3 Sampling: choosing the time instants

Definition: Sampling

Sampling measures a continuous-time signal at selected instants. Uniform sampling uses the same time interval between consecutive measurements.

$$T_s = \frac{1}{f_s}, \quad x[n] = x(nT_s).$$

- f_s : sampling rate in samples per second.
- T_s : time between samples, in seconds.
- n : integer sample number, 0, 1, 2, ...
- $x[n]$: value of the sample taken at time nT_s .

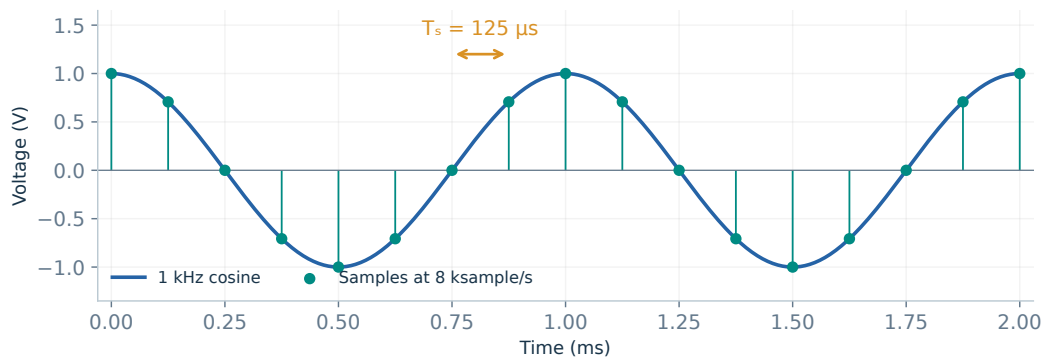


Figure 1.4: The continuous curve is measured at the marked instants.

Example 2

Question

A signal is sampled at 8000 samples/s. Find the time between samples.

Solution

Given: $f_s = 8000$ samples/s. **Required:** T_s .

1. **Choose the rule.** $T_s = 1/f_s$.
2. **Substitute.** $T_s = 1/8000 = 0.000125$ s.
3. **Convert the unit.** $0.000125 \times 10^6 = 125 \mu\text{s}$.

Answer: One sample is taken every $125 \mu\text{s}$.

Remember. Sampling changes the time description. It does not, by itself, round the amplitude to a fixed level.

1.4 Quantization and encoding

Definition: Quantization

Quantization replaces each sample value by one of a finite number of allowed amplitude levels.

Definition: Encoding

Encoding assigns a bit word to the selected level. With b bits per word, there are $L = 2^b$ possible words.

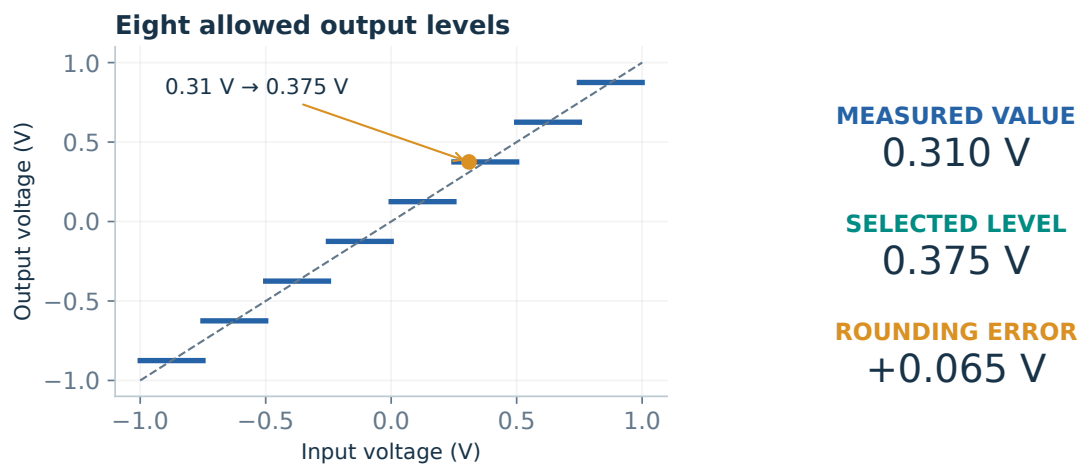


Figure 1.5: Quantization rounds sample values. The difference from the original value is the quantization error.

For a uniform quantizer with range $V_{\min}$ to $V_{\max}$ and L equal bins,

$$\Delta = \frac{V_{\max} - V_{\min}}{L}.$$

Here Δ is the bin width. For midpoint rounding without overload, the error magnitude is at most $\Delta/2$.

Example 3

Question

A uniform 3-bit quantizer covers -1 to $+1$ V. Find the number of levels, bin width and maximum rounding error.

Solution

Given: $b = 3$, $V_{\min} = -1$ V, $V_{\max} = 1$ V.

1. **Count the levels.** $L = 2^3 = 8$.
2. **Divide the voltage range.** $\Delta = [1 - (-1)]/8 = 0.25$ V.
3. **Find half a bin.** $|e_q| \leq 0.25/2 = 0.125$ V.

Answer: 8 levels, bin width 0.25 V, and error magnitude at most 0.125 V for samples within the quantizer range.

1.5 What do the other blocks do?

The basic path is now clear. A practical system can include extra blocks to make transmission more efficient or more reliable. In the diagram, coding prepares the data before transmission. Decoding reverses the representation after the receiver has made its decisions.

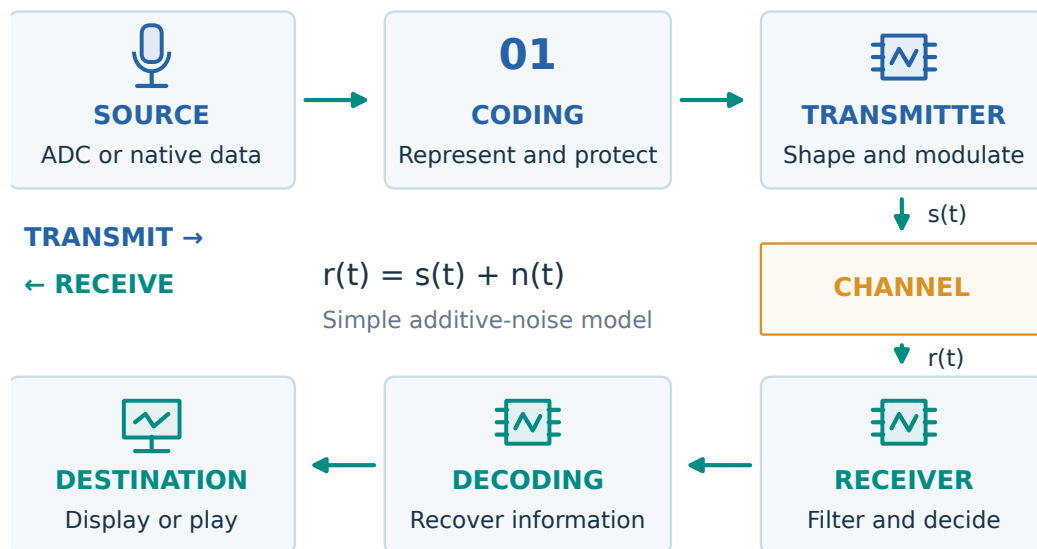


Figure 1.6: A more detailed link. The upper row prepares and transmits information. The lower row receives, decodes and delivers it. ADC means analog-to-digital converter.

- **Source coding:** represents information efficiently, for example by compressing an image.
- **Channel coding:** adds structured extra bits to help detect or correct errors.
- **Pulse shaping:** creates a controlled waveform from the symbols.
- **Carrier modulation:** places information on a carrier when a passband channel is used.
- **Detection:** decides which symbol was most likely sent.

Remember. Source coding and channel coding have different purposes. Compression reduces predictable redundancy. Error-control coding adds useful redundancy. A baseband wired link does not necessarily need radio-frequency (RF) modulation.

Simple example

A receiver observes +0.8 V. The system uses +1 V for bit 1 and −1 V for bit 0. With a zero threshold, the receiver decides **1**, because +0.8 V is positive.

1.6 Communication channels

Definition: Channel

A communication channel is the path through which a transmitted signal reaches a receiver.

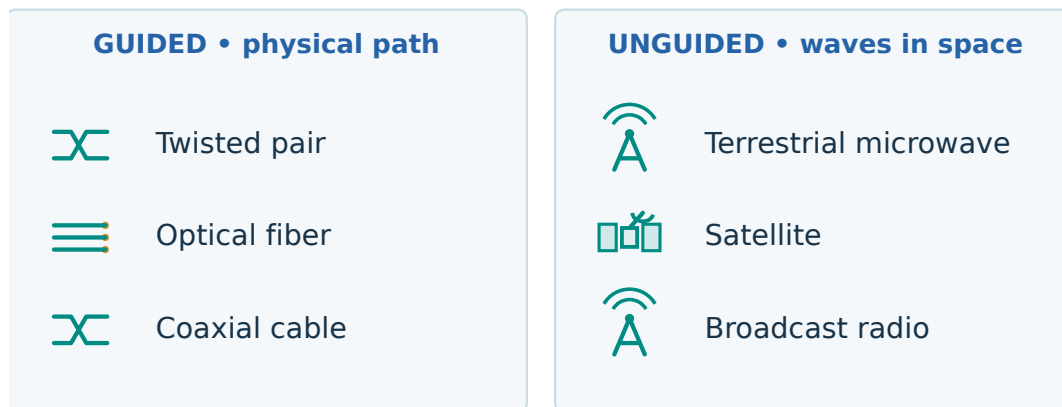


Figure 1.7: Guided channels confine the signal to a physical path. Unguided channels transmit waves through space.

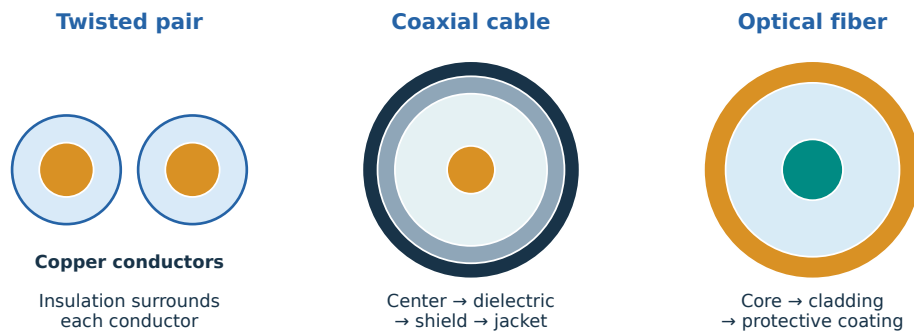
Channel	What carries the signal?	Familiar example
Twisted pair	Electrical signal on a pair of wires	Copper network cable
Coaxial cable	Electrical signal between conductor and shield	Antenna feeder
Optical fiber	Light in a glass fiber	Optical data link
Terrestrial microwave	Radio wave between ground antennas	Point-to-point link
Satellite	Radio wave via a satellite	Satellite television
Broadcast radio	Radio wave to many receivers	Radio broadcasting

Remember. Choose a channel by its required distance, data rate, interference environment and installation conditions. There is no single best medium for every link.

Quick question

Is optical fiber a wireless channel because it carries light? **Answer:** No. The fiber guides the light, so it is a guided channel.

1.7 Guided channels: a closer look



Cross-sections are schematic; dimensions are not to scale.

Figure 1.8: The cross-sections show the layer order in three common media. Dimensions are schematic, not to scale.

1. **Twisted pair.** Two insulated conductors are twisted together. In a balanced link, the receiver measures the voltage difference between them. Twisting helps reduce interference pickup.
2. **Coaxial cable.** A central conductor is surrounded by insulation and an outer conducting shield. The geometry helps confine the electromagnetic field.
3. **Optical fiber.** A core and cladding guide light. The optical signal is not affected by electromagnetic pickup in the way a copper signal is.

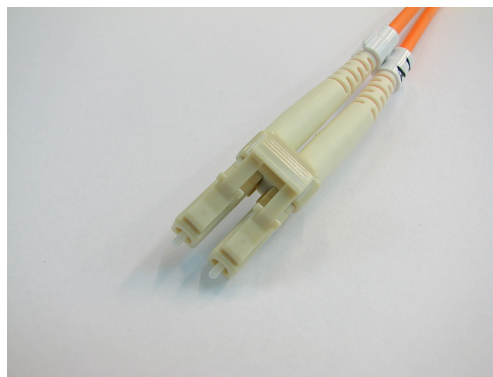


Figure 1.9: A real LC optical-fiber connector. Its ferrule holds the fiber accurately so that two fiber ends can be aligned.

Compare before choosing

Copper connections carry electrical signals and need suitable protection from interference. Fiber is useful where electrical isolation or long-distance optical transmission is needed. Fiber connectors must be clean and correctly aligned.

Remember. Fiber still has losses and signal distortion. Immunity to electrical interference does not mean that every optical link is error-free.

1.8 Wireless channels and signal damage

- **Terrestrial microwave:** directional antennas connect two ground sites.
- **Satellite link:** a satellite relays a signal between ground stations.
- **Broadcast radio:** one transmitter serves many receivers.



Figure 1.10: Dish antennas at the Goldstone complex. A reflector directs radio energy toward the feed when receiving and forms a narrow beam when transmitting.

Any channel can change the transmitted waveform. Three effects are especially important.

Definition: Attenuation

A reduction in signal amplitude or power. The received signal becomes weaker.

Definition: Distortion

A change in waveform shape, for example because different frequency components experience different gains or delays.

Definition: Noise

An unwanted random contribution added to the received signal.

Remember. A weak signal, a distorted signal and a noisy signal are different problems. On the next page, compare their shapes before using any equations.

1.9 Recognizing channel effects

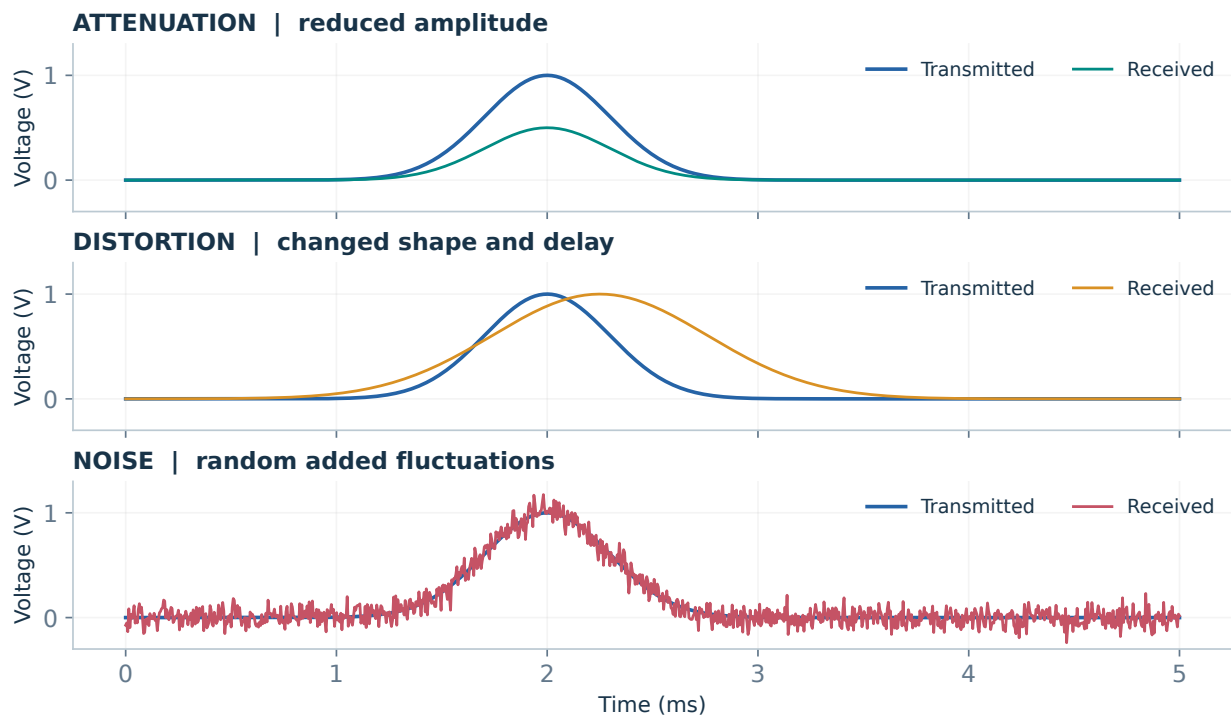


Figure 1.11: Compare each received waveform with the reference: smaller amplitude, changed shape, or random fluctuations.

A simple noise-only model is

$$r(t) = s(t) + n(t).$$

Here $s(t)$ is the transmitted signal, $n(t)$ is additive noise, and $r(t)$ is the received signal. A simple attenuation-and-noise model is $r(t) = as(t) + n(t)$, where $0 < a < 1$ is an amplitude scale factor.

Example 4

Question

A transmitted sample is +1 V. The channel scales it by 0.5 and adds -0.1 V of noise. Find the received sample.

Solution

Given: $s = 1$ V, $a = 0.5$, $n = -0.1$ V.

1. **Apply attenuation.** $as = 0.5 \times 1 = 0.5$ V.
2. **Add the noise.** $r = 0.5 + (-0.1) = 0.4$ V.

Answer: The receiver observes 0.4 V. A zero-threshold binary detector would still decide 1.

1.10 The sampling rule

Definition: Baseband bandwidth

For this lecture, a real baseband signal has bandwidth B when its spectrum has no content outside $-B$ to $+B$. Thus B is its highest positive frequency.

To preserve a band-limited baseband signal, use

$$f_s > 2B.$$

The boundary $2B$ is called the **Nyquist rate**. Its reciprocal $1/(2B)$ is the **Nyquist interval**. Practical sampling uses a rate above the boundary to allow a real filter transition band.

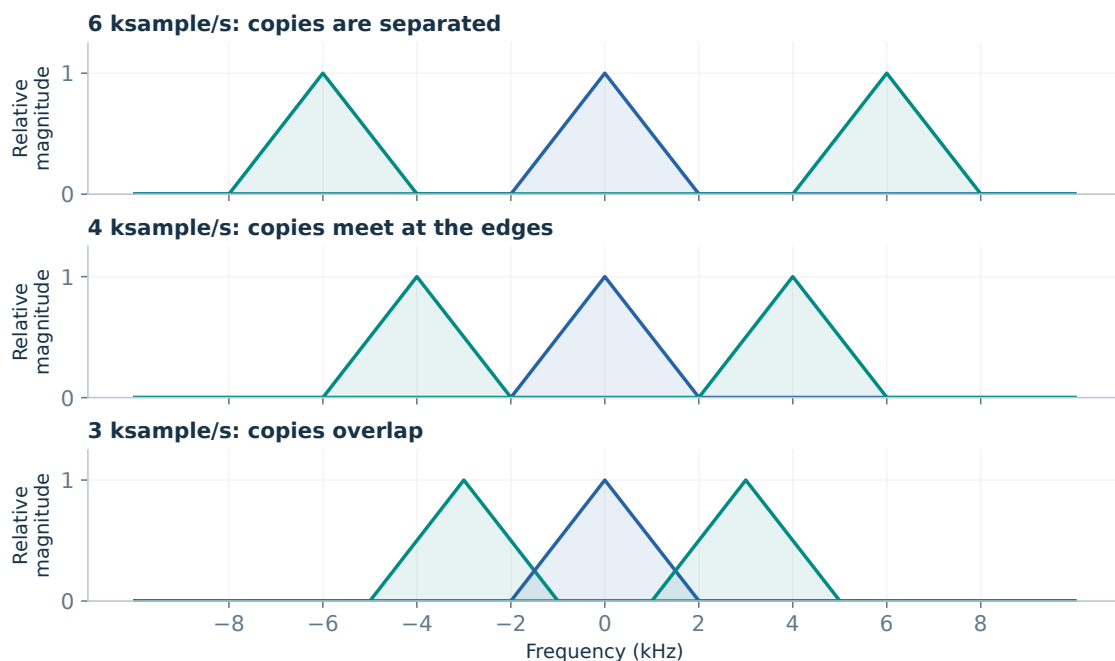


Figure 1.12: For a 2 kHz baseband limit, the repeated spectra are separate, touching, or overlapping. Only the central copy and its two neighbors are shown. Heights are normalized.

Remember. The equality $f_s = 2B$ is a limiting case, not a safe choice for every signal. A sine wave exactly at the boundary can produce all-zero samples for a particular phase.

Three cases

$f_s > 2B$: separated copies. $f_s = 2B$: copies meet at the boundary.

$f_s < 2B$: copies overlap for a general signal occupying the band.

1.11 Finding a frequency from an equation

A sinusoid is commonly written as

$$x(t) = A \cos(2\pi ft + \phi).$$

A is amplitude, f is frequency in hertz, t is time in seconds, and ϕ is phase in radians. If the equation uses angular frequency ω , then $\omega = 2\pi f$.

Example 5

Question

For $x(t) = 8 \cos(200\pi t)$, find the signal frequency, Nyquist rate and Nyquist interval.

Solution

Given: The coefficient of t is 200π rad/s.

1. **Compare with the standard form.** $2\pi f = 200\pi$.
2. **Solve for frequency.** $f = 200\pi / (2\pi) = 100$ Hz. This single-tone signal has $B = 100$ Hz.
3. **Double the highest frequency.** $R_N = 2B = 2(100) = 200$ samples/s.
4. **Take the reciprocal.** $T_N = 1/R_N = 1/200 = 0.005$ s = 5 ms.

Answer: Frequency: 100 Hz. Nyquist rate: 200 samples/s. Nyquist interval: 5 ms. Choose an actual sampling rate above 200 samples/s.

Remember. Do not use 200π as a frequency in hertz. Divide the angular frequency by 2π first. Amplitude 8 does not change the Nyquist rate.

Check yourself

For $x(t) = 3 \sin(600\pi t)$, $f = 300$ Hz and the Nyquist rate is 600 samples/s. The amplitude 3 and the use of sine instead of cosine do not change this calculation.

1.12 A sum of sinusoids

When several different sinusoidal components are added, list their frequencies and select the largest one that remains in the signal. Work with the frequency inside each sine or cosine. The number multiplying the whole term is its amplitude.

Example 6

Question

Find the Nyquist rate for $x(t) = 3 \cos(50\pi t) + 10 \sin(300\pi t) - \cos(100\pi t)$.

Solution

Given: Three components with angular frequencies 50π , 300π and 100π rad/s.

1. **Convert each frequency.** Divide each coefficient of t by 2π .

Component	Calculation	Frequency
$3 \cos(50\pi t)$	$50\pi / (2\pi)$	25 Hz
$10 \sin(300\pi t)$	$300\pi / (2\pi)$	150 Hz
$-\cos(100\pi t)$	$100\pi / (2\pi)$	50 Hz

2. **Choose the largest.** $B = \max(25, 150, 50) = 150$ Hz.
3. **Apply the sampling rule.** $R_N = 2B = 2(150) = 300$ samples/s.

Answer: The Nyquist rate is 300 samples/s. Use an actual rate greater than 300 samples/s for this baseband signal.

Remember. Do not add the component frequencies. The highest frequency sets the baseband limit. The minus sign changes a component's sign, not its frequency.

Why does this method work?

Sampling must preserve the fastest oscillation present. A rate that preserves the 150 Hz component also exceeds the boundary for the 25 Hz and 50 Hz components.

1.13 A product of sinusoids

Multiplication creates new frequencies. Expand a product before finding its highest frequency. The sum-frequency term oscillates faster than either original component, so it determines the required sampling rate in this example.

Example 7

Question

Find the Nyquist rate and interval for $x(t) = \cos(4000\pi t) \cos(1000\pi t)$.

Solution

Given: The two original frequencies are 2000 Hz and 500 Hz.

1. **Write the identity.** $\cos A \cos B = \frac{1}{2} \cos(A + B) + \frac{1}{2} \cos(A - B)$.

2. **Expand the product.**

$$x(t) = \frac{1}{2} \cos(5000\pi t) + \frac{1}{2} \cos(3000\pi t).$$

3. **Read the new frequencies.** $f_1 = 2500$ Hz and $f_2 = 1500$ Hz.

4. **Find the rate.** $R_N = 2(2500) = 5000$ samples/s.

5. **Find the interval.** $T_N = 1/5000 = 0.0002$ s = 0.2 ms.

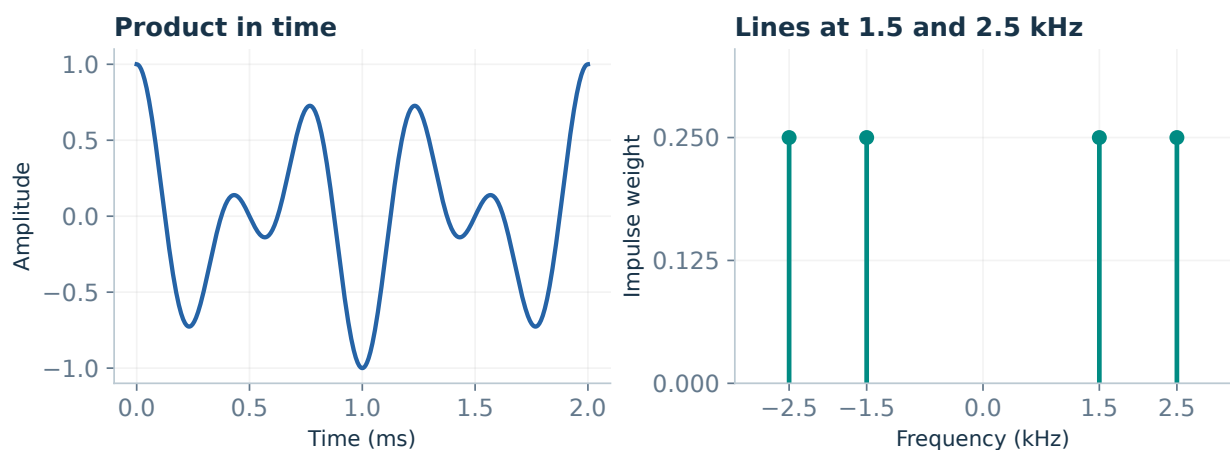


Figure 1.13: Multiplication produces components at the sum and difference frequencies. Each bilateral Fourier impulse has weight 1/4.

Answer: Nyquist rate: 5000 samples/s. Nyquist interval: 0.2 ms.

Remember. For a product, choosing the larger original frequency would give the wrong answer. Expand first.

1.14 Aliasing: a false frequency

Definition: Aliasing

Aliasing occurs when different continuous-time signals give the same samples. The samples alone cannot distinguish those signals.

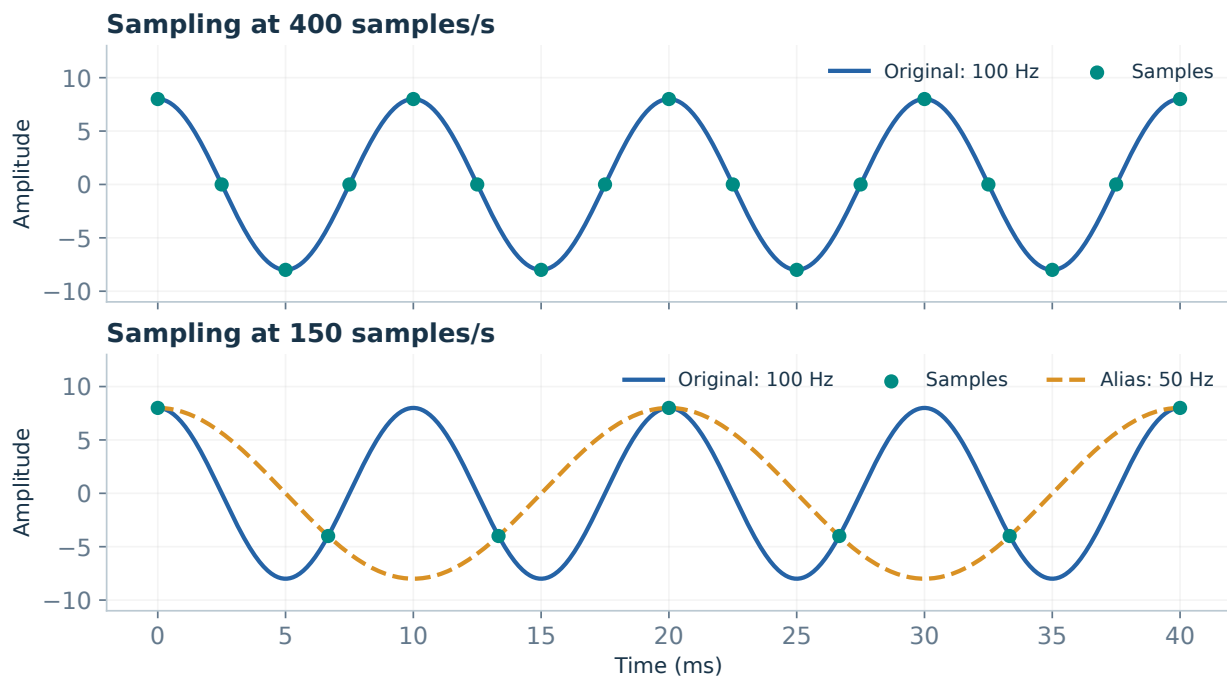


Figure 1.14: At 150 samples/s, the 100 Hz and 50 Hz cosines have the same sample values.

Example 8

Question

A 100 Hz cosine is sampled at 150 samples/s. Find the alias frequency between 0 and $f_s/2$.

Solution

Given: $f = 100$ Hz, $f_s = 150$ samples/s. The interval ends at 75 Hz.

1. **Check the sampling rate.** $150 < 2(100)$, so this rate is below the Nyquist rate.
2. **Find a frequency in the first interval.** $f_a = |f - f_s| = |100 - 150| = 50$ Hz.
3. **Check the interval.** $0 \leq 50 \leq 75$ Hz.

Answer: The samples are also consistent with a 50 Hz cosine.

Remember. For a single cosine, choose an integer k so that $f_a = |f - kf_s|$ lies between 0 and $f_s/2$. One subtraction is not enough for every frequency.

1.15 Writing the sampled signal

To find the sample sequence, replace time t by n/f_s . The sequence is a function of the integer n . For example, $n = 2$ means the measurement at $2/f_s$ seconds. The coefficient of n is an angle per sample, not a frequency in hertz.

Example 9

Question

For $x(t) = 8 \cos(200\pi t)$, find $x[n]$ at (a) 400 samples/s and (b) 150 samples/s.

Solution

Given: $f = 100$ Hz. Use $x[n] = x(n/f_s)$.

(a) Sampling at 400 samples/s

1. **Substitute the sample time.** $x[n] = 8 \cos(200\pi n/400)$.
2. **Simplify.** $x[n] = 8 \cos(\pi n/2)$.
3. **Check the first values.** For $n = 0, 1, 2, 3$, the values are 8, 0, -8, 0. There are four samples per cycle.

(b) Sampling at 150 samples/s

1. **Substitute the sample time.** $x[n] = 8 \cos(4\pi n/3)$.
2. **Use periodicity for integer n .** Subtract $2\pi n$ inside the cosine: $4\pi n/3 - 2\pi n = -2\pi n/3$.
3. **Use cosine symmetry.** $\cos(-\theta) = \cos \theta$, so $x[n] = 8 \cos(2\pi n/3)$.
4. **Identify the alias.** $2\pi f_a/150 = 2\pi/3$, giving $f_a = 50$ Hz.

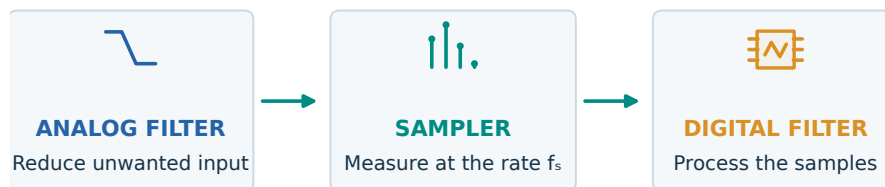
Answer: At 400 samples/s: $8 \cos(\pi n/2)$. At 150 samples/s: $8 \cos(2\pi n/3)$, which also represents a sampled 50 Hz cosine.

Remember. The expressions $8 \cos(4\pi n/3)$ and $8 \cos(2\pi n/3)$ agree at integer n . They are not identical for arbitrary real-valued time.

1.16 Preventing aliasing and recovering a signal

Definition: Anti-aliasing filter

An analog filter placed before the sampler to reduce frequency components that would fold into the wanted band.



6 kHz sampled at 8 ksample/s aliases to 2 kHz: filter BEFORE sampling.

Figure 1.15: The unwanted component must be reduced before sampling.

A digital filter cannot separate two components that have already become indistinguishable through aliasing. The protection therefore starts in the analog path.

Definition: Reconstruction

Reconstruction forms a continuous-time signal from its samples. Ideal reconstruction of a band-limited signal uses a low-pass interpolation filter.

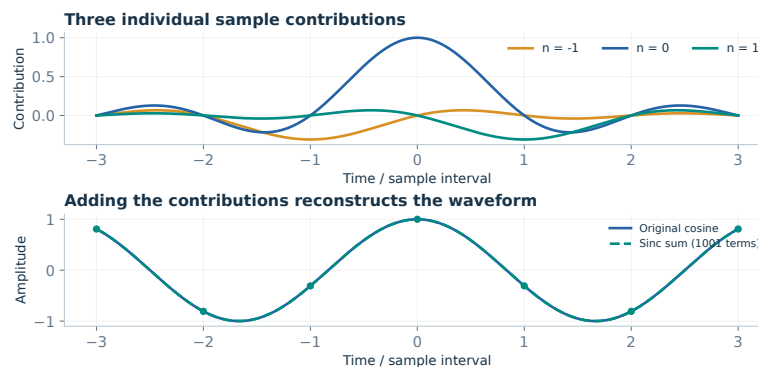


Figure 1.16: Three individual interpolation pulses are shown above. Below, a 1001-term sinc sum closely matches the original cosine.

The practical sequence

1. Limit the input bandwidth with an analog filter.
2. Sample fast enough for the filtered signal.
3. Process or transmit the sample values.
4. Use a reconstruction stage to form the output waveform.

1.17 Why does the spectrum repeat?

Detailed explanation after the basic rule

A spectrum describes which frequencies are present. Let $X(f)$ be the Fourier transform of $x(t)$, using frequency f in hertz.

1. **Mark every sampling instant.** An ideal impulse train is

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s).$$

The impulse $\delta(t - nT_s)$ marks the instant nT_s .

2. **Select the signal values.** Multiply by the impulse train:

$$x_s(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s).$$

The weight of each impulse is the corresponding sample value.

3. **Use the impulse-train transform.** Its transform is another impulse train:

$$P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(f - kf_s).$$

The integer k identifies a spectral copy. For $k = 0$, the copy stays at zero. For $k = 1$, its center moves to f_s . For $k = -1$, its center moves to $-f_s$.

4. **Apply multiplication–convolution duality.** Multiplication in time becomes convolution in frequency:

$$X_s(f) = X(f) * P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(f - kf_s).$$

Here $*$ denotes convolution. Each term is a shifted copy of $X(f)$.

5. **Require a gap between copies.** The original ends at B . The next copy starts at $f_s - B$. Separation requires

$$B < f_s - B \implies f_s > 2B.$$

Remember. The algebra leads to the same picture shown earlier: copies are spaced by f_s , and each baseband copy extends from $-B$ to B .

1.18 The reconstruction formula

The reconstruction filter keeps the original spectral copy and rejects the others. Its gain also removes the scale factor $1/T_s$ introduced by sampling.

1. **Choose an ideal low-pass filter.** For $f_s > 2B$, choose

$$H(f) = \begin{cases} T_s, & |f| < f_s/2, \\ 0, & |f| > f_s/2. \end{cases}$$

Inside the original band, $T_s X_s(f) = X(f)$.

2. **Find its impulse response.** The inverse Fourier transform gives

$$h(t) = T_s \int_{-f_s/2}^{f_s/2} e^{j2\pi ft} df = T_s \frac{\sin(\pi f_s t)}{\pi t} = \text{sinc}(t/T_s).$$

Here $j^2 = -1$, $f_s T_s = 1$, and $\text{sinc}(u) = \sin(\pi u)/(\pi u)$ with $\text{sinc}(0) = 1$.

3. **Add the response to every sample impulse.** Convolution $x_s(t)$ with $h(t)$ gives

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t - nT_s}{T_s}\right).$$

Each sample sets the height of one shifted sinc pulse.

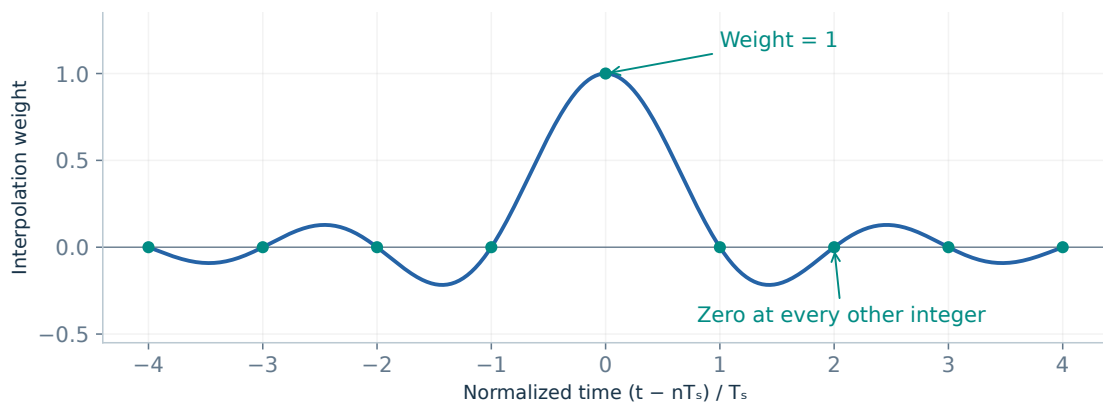


Figure 1.17: A sinc pulse is 1 at its own sample instant and 0 at the other integer sample positions.

Remember. At $t = mT_s$, only the term $n = m$ remains, so the formula returns $x[m]$. The ideal formula uses infinitely many samples. Practical filters approximate it.

1.19 A squared-sinc example

Example 10

Question

For $x(t) = \text{sinc}^2(200t)$, find the baseband bandwidth, Nyquist rate and Nyquist interval. Use normalized sinc.

Solution

Given: $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.

1. **Start with the unsquared sinc.** The transform of $\text{sinc}(200t)$ is a rectangle extending from -100 to $+100$ Hz. Its scale factor is $1/200$.
2. **Apply the square.** Squaring in time convolves the rectangle with itself in frequency.
3. **Add the frequency limits.** The new support extends from $-100 - 100 = -200$ Hz to $100 + 100 = 200$ Hz. Its shape is triangular.
4. **Read the baseband bandwidth.** $B = 200$ Hz, the positive upper edge.
5. **Calculate rate and interval.** $R_N = 2B = 400$ samples/s and $T_N = 1/400 = 2.5$ ms.

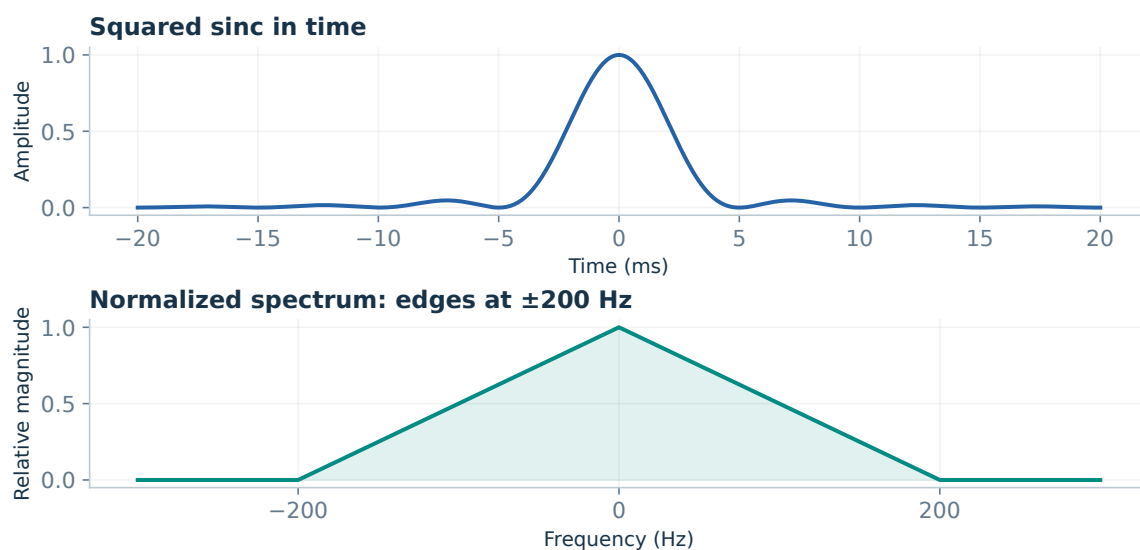


Figure 1.18: The squared-sinc waveform has a triangular spectrum with edges at ± 200 Hz.

Answer: Bandwidth: 200 Hz. Nyquist rate: 400 samples/s. Nyquist interval: 2.5 ms.

Remember. The total spectral width is 400 Hz, but the baseband bandwidth is 200 Hz. Squaring the waveform does not mean simply doubling its time argument.

1.20 Bandpass sampling

Definition: Bandpass signal

A bandpass signal occupies a band away from zero frequency. Write its positive-frequency edges as f_L and f_H . Its occupied bandwidth is $B = f_H - f_L$.

Sampling above $2f_H$ is a straightforward safe choice. A lower rate may also work if the shifted positive and negative bands do not overlap.

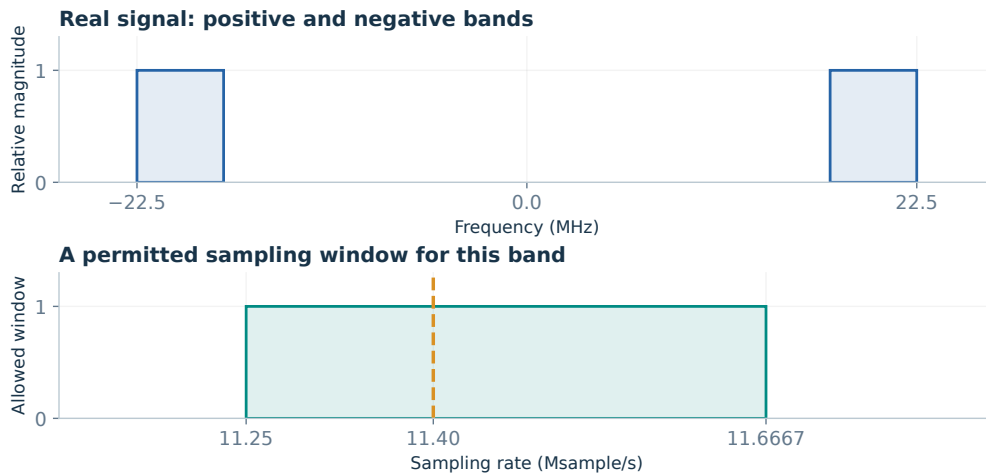


Figure 1.19: An occupied band away from zero can fit into a suitable sampling window.

How to test a lower sampling rate

For an ideal band-limited real signal, a candidate window is

$$\frac{2f_H}{m} \leq f_s \leq \frac{2f_L}{m-1}, \quad 2 \leq m \leq \left\lfloor \frac{f_H}{B} \right\rfloor.$$

- m is an integer that selects a possible sampling window.
- $\lfloor \cdot \rfloor$ means round down to an integer.
- Calculate both ends of the window before choosing f_s .
- Use an interior value in practice, allowing filter and frequency margins.

Remember. Do not use $f_s = 2B$ automatically for a bandpass signal. The band location matters as well as its width. If no $m \geq 2$ is available, use the ordinary rate above $2f_H$.

1.21 A bandpass sampling example

Example 11

Question

A real signal occupies 17.5 to 22.5 MHz on its positive-frequency side. Find a permitted lower sampling-rate window and select a rate inside it.

Solution

Given: $f_L = 17.5$ MHz and $f_H = 22.5$ MHz.

1. **Calculate the occupied bandwidth.** $B = 22.5 - 17.5 = 5$ MHz.
2. **Find the largest permitted integer.** $\lfloor f_H/B \rfloor = \lfloor 22.5/5 \rfloor = 4$. Choose $m = 4$ to test the lowest candidate window.
3. **Calculate the lower limit.** $2f_H/m = 2(22.5)/4 = 11.25$ MHz.
4. **Calculate the upper limit.** $2f_L/(m-1) = 2(17.5)/3 = 11.6667$ MHz.
5. **Choose an interior rate.** Use $f_s = 11.4$ Msample/s, which lies between the two limits.

Answer: A candidate window is $11.25 \leq f_s \leq 11.6667$ Msample/s. The choice 11.4 Msample/s is inside it.

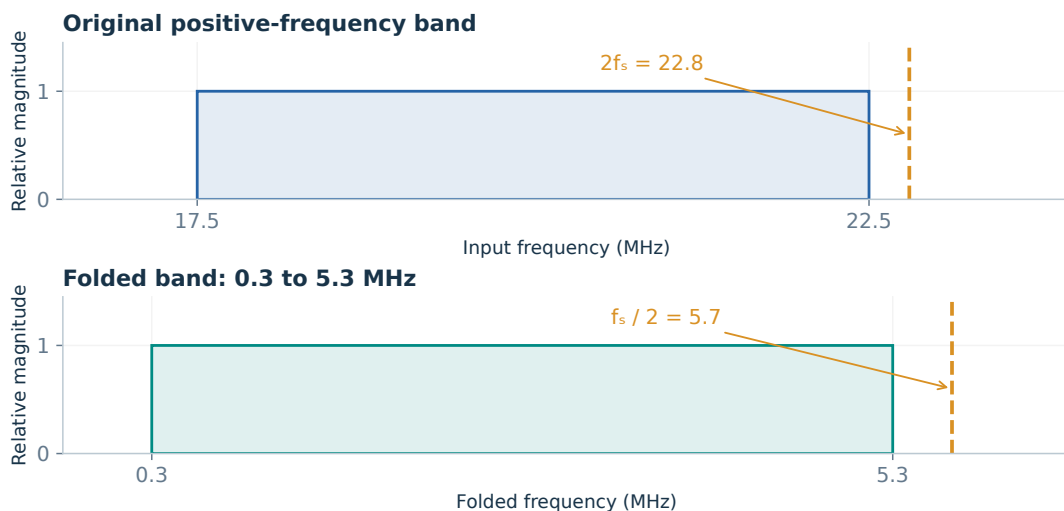


Figure 1.20: At 11.4 Msample/s, the positive RF band folds into 0.3–5.3 MHz, inside the interval 0–5.7 MHz.

Check the folded limits

$2f_s - f_H = 22.8 - 22.5 = 0.3$ MHz and $2f_s - f_L = 22.8 - 17.5 = 5.3$ MHz. Both lie below $f_s/2 = 5.7$ MHz. The order reverses, so the folded band is spectrally inverted. This means that the original upper edge becomes the lower folded edge. A receiver must account for this reversal when interpreting the spectrum.

1.22 Practice with short solutions

Example 12

Question

A baseband signal has bandwidth 3 kHz. Find the Nyquist rate. If sampled at 8 ksample/s, find the sample interval.

Solution

Given: $B = 3000$ Hz and $f_s = 8000$ samples/s.

1. **Find the boundary.** $R_N = 2B = 6000$ samples/s.
2. **Check the chosen rate.** $8000 > 6000$, so it exceeds the boundary.
3. **Find the interval.** $T_s = 1/8000 = 125 \mu\text{s}$.

Answer: Nyquist rate: 6 ksample/s. Actual sample spacing: $125 \mu\text{s}$.

Example 13

Question

A 900 Hz cosine is sampled at 1200 samples/s. Find its alias in the interval 0–600 Hz.

Solution

Given: $f = 900$ Hz and $f_s = 1200$ samples/s.

1. **Check the boundary.** $1200 < 2(900)$, so undersampling occurs.
2. **Fold the frequency.** $f_a = |900 - 1200| = 300$ Hz.
3. **Verify the interval.** $300 < 600$ Hz.

Answer: The alias frequency is 300 Hz.

Before accepting any answer

1. Did you convert radians per second to hertz?
2. Did you find the largest frequency *after* expanding any product?
3. Did you distinguish the Nyquist rate from the actual sampling rate?
4. Are your units seconds, milliseconds or microseconds?

1.23 Chapter summary and reading

Idea	Meaning or rule
Sampling	Choose when to measure: $x[n] = x(nT_s)$
Quantization	Choose an allowed amplitude level
Encoding	Assign a bit word to the chosen level
Sample interval	$T_s = 1/f_s$
Sinusoidal frequency	$f = \omega/(2\pi)$
Baseband Nyquist rate	$R_N = 2B$
Baseband Nyquist interval	$T_N = 1/(2B)$
Practical baseband choice	$f_s > 2B$, with filter margin
Single-cosine alias	$ f - kf_s $ in $[0, f_s/2]$
Reconstruction	Recover a waveform from its samples

Explain these in your own words

Why are sampling and quantization different? Why can a digital link carry a continuous waveform? Why does aliasing prevent unique recovery? Why should a filter come before the sampler?

Further reading

- [1] S. Sharma, *Principles of Communication*, S. K. Kataria & Sons. Consult the digital communication and sampling chapters in the assigned edition.
- [2] J. G. Proakis and M. Salehi, *Digital Communications*, McGraw-Hill. Further reading on digital communication systems and signal representation.

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1.24 Homework

Show the rule, substitute the values, keep the units, and write a final sentence. For sampling questions, distinguish a theoretical Nyquist rate from a practical choice above it.

A. Understand the system

- H1. Draw the five main blocks of a digital communication system. Write one short sentence explaining each block.
- H2. Explain sampling, quantization and encoding using one measured voltage as an example.
- H3. Classify twisted pair, optical fiber, satellite and broadcast radio as guided or unguided.
- H4. Sketch one clean pulse and three received versions showing attenuation, distortion and noise.

B. Use the basic rules

- H5. A signal is sampled at 10 ksample/s. Find T_s in seconds and microseconds.
- H6. A uniform 4-bit midpoint quantizer covers -2 to $+2$ V. Find the number of levels, bin width and maximum rounding-error magnitude without overload.
- H7. For $x(t) = 5\cos(800\pi t)$, find frequency, Nyquist rate and Nyquist interval. Choose a suitable rate above the boundary.
- H8. Find the Nyquist rate for $x(t) = 2\sin(100\pi t) + 4\cos(600\pi t)$. Explain why amplitude does not set this rate.
- H9. Expand $x(t) = \cos(2000\pi t)\cos(600\pi t)$. Find the two frequencies, Nyquist rate and Nyquist interval.
- H10. A 700 Hz cosine is sampled at 1000 samples/s. Find the alias in 0–500 Hz.
- H11. Write the sample sequence of $x(t) = 2\cos(200\pi t)$ at 800 samples/s. Calculate its first four sample values.

C. Extend your understanding

- H12. Find the bandwidth and Nyquist rate of $\text{sinc}^2(100t)$ using normalized sinc. Sketch its spectral support.
- H13. A real bandpass signal occupies 9–11 MHz. Test all allowed integers $m \geq 2$, list the candidate sampling windows, and choose one interior rate.
- H14. Explain why a 6 kHz unwanted cosine can appear at 2 kHz after sampling at 8 ksample/s. Where should the filter be placed?